

INTRODUCTION TO MACROECONOMICS – June 2021

Q1

Among the following propositions, which one describes an issue analyzed by macroeconomics?

- (a) The location choice of a Swiss company needing to choose between setting up its productive units in Switzerland or in Bulgaria, where labor costs are lower than in Switzerland.
- (b) The investment decision of an investor who must choose between purchasing a domestic security or a foreign security, knowing that the interest attached to the foreign security is higher than that attached to the domestic security, but that the risk is also higher.
- (c) The effect on companies' investment expenditures caused by an expansionary fiscal policy which provokes an increase in public debt at 110% of GDP.
- (d) The impact on its reserves of the decision by a commercial bank to grant a new mortgage to a real estate developer.

Correct answer: (c)

Q2

Which of the following will **NOT** be included while calculating the gross domestic product (GDP) of Switzerland?

- (a) The production of watches by a French employee in a Swiss firm.
- (b) The production of chocolate in Switzerland using cocoa beans imported from Colombia.
- (c) The production of precision instruments by a Swiss firm located in France.
- (d) The production of wine by a foreign-owned firm located in Switzerland.

Correct answer: (c)

Q3

Suppose that country A measures GDP using both the income approach and the expenditure approach. Country A's consumption is 200, investment is 100, government expenditure is 300 and net exports is 100. Capital income is 200 and entrepreneur profits is 100. What is the labor income in the economy?

- (a) 700.
- (b) 500.
- (c) 300.
- (d) 400.

Correct answer: (d)

Q4

Suppose that your income increases from 20'000 CHF to 25'000 CHF while, simultaneously, the Consumer Price Index (CPI) increases from 120 to 150. Your purchasing power:

- (a) remains constant.
- (b) increases.

- (c) cannot answer this question without knowing the base year.
- (d) cannot answer this question without knowing the time interval.

Correct answer: (a)

Q5

Let's suppose there is a consumer price index based on an annual study of expenditures spent by a representative consumer in Geneva. This price index is based on the following table for the year 2015 (base year) and 2020.

Good	Quantity in 2015	Quantity in 2020	Unit price in 2015 (CHF)	Unit price in 2020 (CHF)
Milk	10	15	8	15
Clothes	10	10	27	18
Meat	20	15	20	30
Coffee	20	18	2	3
Mobile subscription	2	6	5	5

Among the following propositions, which one is correct? The Consumer Price Index for Geneva is:

- (a) 80.
- (b) 125.
- (c) 124.
- (d) 117.

Correct answer: (b)

Q6

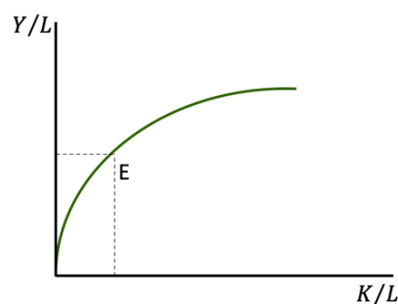
Among the following elements, which one could have caused an overestimation of the cost of living by the Consumer Price Index?

- (a) Following an increase in travel costs by train, consumers favored travelling by car.
- (b) A generalized increase in prices caused an inflation rate increase by 1%.
- (c) In order to account for changes in consumers' feeding habits, the composition of the consumption basket of the average consumer has been modified and the share attributed to meat consumption has been decreased.
- (d) Telecommunications, whose prices have gradually increased, represent an increasingly important part of total household expenditure.

Correct answer: (a)

Q7

Suppose that Moneyland displays the following production function: $Y/L = A \cdot F(1, K/L, H/L, N/L)$, with Y/L being the output per labor, A the total factor productivity, K/L the capital per labor, H/L the human capital per labor, N/L the natural resources per labor. Consider the graph below representing the production function of Moneyland and suppose that the economy is initially in equilibrium at point E:



Among the following propositions, which one is correct?

- (a) In case of an increase in human capital, the new equilibrium output will move to the right along the production function.
- (b) The discovery of new natural resources in Moneyland shifts the production function up.
- (c) Moneyland's production function displays decreasing returns to scale.
- (d) A change in total factor productivity will have no impact on the output per labor as the capital to labor ratio is not affected and remains constant.

Correct answer: (b)

Q8

Among the following propositions regarding economic convergence, which one is correct?

- (a) If country A has a GDP per capita 10 times higher than country B, because of production factors' decreasing marginal productivity, the growth rate of GDP per capita in country B should theoretically be larger than that of country A. Eventually, the growth rate of the two countries should be the same.
- (b) According to the predictions of economic theory, poor countries will not be able to catch up with richer countries because of a constant lack of physical capital.
- (c) Conditional economic convergence can explain the gradual catching-up of new emerging economies, like China and India, with living standards of industrialized countries.
- (d) If country A has a GDP per capita 10 times higher than country B, because of production factors' increasing marginal productivity, the growth rate of GDP per capita in country B should theoretically be lower than that of country A. Eventually, the differential in GDP per capita between countries A and B should be strengthened.

Correct answer: (a)

Q9

Suppose that the labor market in Petroland is initially at full employment in equilibrium and that only oil is produced in this economy. If the world oil price decreases, among the following propositions which one is correct considering that wages are fixed over a relatively long period?

- (a) Petroland's economy remains at full employment.
- (b) The reservation wage changes for all the workers in Petroland.
- (c) Petroland's equilibrium wage decreases.
- (d) Petroland's labor demand is lower than labor supply.

Correct answer: (d)

Q10

You work as an economist for the U.S Bureau of Labor Statistics and you have the following indicators on the U.S economy in March 2021:

Total population	333 million
Working age population	200 million
Unemployment rate	6%
Labor force participation rate	60%

Your chief economist asks you to write a report on the latest trends in the U.S labor force. In your report, you present the following statistics, respectively, for the U.S active workforce and the total number of workers:

- (a) 120 million ; 112.8 million.
- (b) 200 million ; 188 million.
- (c) 120 million ; 7.2 million.
- (d) 200 million; 12 million.

Correct answer: (a)

Q11

Among the following events, which one can induce structural unemployment?

- (a) The government introduces a minimum wage at a level below the equilibrium market wage.
- (b) Firms set wages at a higher level compared to the equilibrium market wage to attract more productive workers.
- (c) The government's labor agency becomes more inefficient which increases the worker/employer "matching" time.
- (d) The government puts in place an expansionary fiscal policy.

Correct answer: (b)

Q12

Suppose that the interest rate on bonds increases. Among the following propositions, which one is correct?

- (a) People will be willing to reduce their liquidity holdings.
- (b) The face value of bonds will decrease.
- (c) People will be willing to increase their liquidity holdings.
- (d) The opportunity cost of holding money decreases.

Correct answer: (a)

Q13

In a closed economy, a tax decrease on interest income and an investment tax credit will have a(n) _____ effect on the interest rate and _____ on loanable funds.

- (a) positive, uncertain.
- (b) negative, positive.
- (c) positive, negative.
- (d) uncertain, positive.

Correct answer: (d)

Q14

Among the following statements, which one/ones is/are true?

- i. Increasing the refinancing rate decreases the money supply as there is less demand for refinancing by commercial banks.
- ii. The refinancing rate is the interest rate at which the Central Bank lends on a short-term basis to the banking sector.
- iii. The monetary multiplier is given by the inverse of the refinancing rate.

- (a) Only i. and iii. are true.
- (b) Only ii. is true.
- (c) Only i. and ii. are true.
- (d) Only iii. is true.

Correct answer: (c)

Q15

Among the following statements which one is correct? During an episode of hyperinflation:

- (a) the quantitative equation is not at work as the velocity of money increases proportionally to money supply.
- (b) money demand is perfectly elastic as uncertainty is high and agents hold huge amounts of money.
- (c) the Fisher effect is broken as nominal interest rates cannot increase indefinitely.
- (d) the inflation tax burden is very high as the purchasing power of money decreases rapidly.

Correct answer: (d)

Q16

Among the following statements regarding the balance of payments, which one is correct?

- (a) The capital account (or financial account) of the balance of payments records the total stock of foreign assets owned by the domestic country,
- (b) Compensation received by a domestic resident as a result of an accident abroad will be recorded in the financial part of the balance of payments.
- (c) A net capital and labor exporting country will have a negative net factor income from abroad.
- (d) Assuming that net factor income from abroad is in equilibrium and that capital transfers are zero, a trade surplus will be compensated by positive net capital outflows.

Correct answer: (d)

Q17

In the long run, the relative Purchasing Power Parity theory predicts the changes in the nominal exchange rate as a function of inflation rate differentials. If price levels in the UK increase at an annual rate of 3% and decrease at an annual rate of 1% in USA, we can conclude that there will be:

- (a) a 1% depreciation of the USD with respect to the GBP.
- (b) a 2% depreciation of the GBP with respect to the USD.
- (c) a 4% appreciation of the USD with respect to the GBP.
- (d) a 3% appreciation of the GBP with respect to the USD.

Correct answer: (b)

Q18

Take Switzerland as the reference country and suppose that you have the following information:

	Price of a Big Mac in local currency	Exchange rate observed on the market
Switzerland	CHF 6	-
Norway	NOK 60	NOK 10 = CHF 1
Sweden	SEK 30	SEK 10 = CHF 1
Israel	NIS 20	NIS 4 = CHF 1

Based on the BigMac index, what can you state?

- (a) The Swedish krona (SEK) is overvalued with respect to the Swiss franc (CHF).
- (b) The Norwegian krone (NOK) and the New Israeli Shekel (NIS) are correctly valued with respect to the Swiss franc (CHF).
- (c) We can expect an appreciation of the New Israeli Shekel (NIS) with respect to the Swiss franc (CHF).
- (d) We can expect an appreciation of the Swiss franc (CHF) with respect to the Swedish krona (SEK).

Correct answer: (c)

Q19

Consider a world consisting in two countries, Uniland and Confeland, which trade together. Currencies of these two countries are the Unido (UND) and the Consol (CNS), respectively. After a period of strong political instability in Confeland, net capital outflows from Confeland increase. Given that Uniland and Confeland are in a floating exchange regime, among the following propositions, which one describes the impact on the foreign exchange market?

- (a) Demand for UND shifts to the right and the CNS depreciates.
- (b) Demand for UND and supply of CNS shift to the right proportionally and the nominal exchange rate between the two currencies does not change.
- (c) Supply of CNS shifts to the left and the CNS appreciates.
- (d) If we do not know which is the domestic country, we cannot state anything regarding the impact of that shock on the nominal exchange rate.

Correct answer: (a)

Q20

Assume that the government of an open economy increases spending by borrowing in the market and, at the same time, increases tax on income from interest payments. Using the theory of the markets of loanable funds, net capital outflow, and foreign currency exchange, what will be the effect of these two events on the equilibrium?

- (a) Loanable funds and net capital outflows will decrease while there will be no impact on the exchange of goods and services.
- (b) Trade balance will deteriorate while the impact on the real interest rate of the country will be ambiguous.
- (c) The currency of the country will appreciate in real terms and its real interest rate will increase.
- (d) There will be a sudden increase in capital outflows that will generate a depreciation of the domestic currency in real terms and a consequent improvement of net exports.

Correct answer: (c)

Q21

The trade balance of an open economy is negative. Given that its GDP is 100, investments are 20 and consumption is 50, among the following statements, which one is correct?

- (a) Public spending is 30.
- (b) We do not have enough information to determine the value of public spending. The only thing we can say is that it is above 30.
- (c) If public savings are 10, private savings are 15.
- (d) We do not have enough information to determine the value of national savings. The only thing we can say is that it is above 20.

Correct answer: (b)

Q22

Suppose that the government of Openland decides to ease its immigration policy, leading to a large inflow of immigrants which increases both the overall population of the country (and thus total consumption), as well as its working population. Based on the AD-AS model and assuming the economy was initially at its long-term equilibrium (i.e., aggregate output is at its natural level), among the following propositions which one is correct?

- (a) In the long-term, output increases and the effect on prices is uncertain.
- (b) The output increases only in the short run: after an adjustment period it goes back to its initial level.
- (c) Prices and output decrease in the short run. In the long-term the output goes back to its initial level and prices increase more than in the initial equilibrium only if the government puts in place an expansionary fiscal policy that boosts demand.
- (d) Both in the short and in the long terms, the effect on output is uncertain.

Correct answer: (a)

Q23

Suppose that the government of Spendyland recently issued its budget for 2022 and announced that public spending will increase by 2.5 billion compared to 2021, while taxes will remain constant. If the marginal propensity to saving (mps) in Spendyland is 0.2, by how much should the GDP increase next year all other things being equal?

- (a) If Spendyland is a closed economy, its GDP will increase by 5 billion.
- (b) If Spendyland is an open economy and the marginal propensity to import (mpi) is 0.3, its GDP will increase by 5 billion.
- (c) If Spendyland is a closed economy, its GDP will increase by 3.125 billion.
- (d) If Spendyland is an open economy and the marginal propensity to import (mpi) is 0.1, its GDP will increase by 25 billion.

Correct answer: (b)

Q24

Among the following propositions which one is correct?

- (a) The Keynesian multiplier of government expenditure is affected by a change in a lump-sum tax.
- (b) The crowding out effect is an economic theory arguing that rising public sector spending drives up private sector spending.
- (c) In the case of a severe recession, automatic stabilizers do not increase aggregate demand unless they are combined with an active fiscal policy.
- (d) None of the other propositions is correct.

Correct answer: (d)

Q25

Assume the overall price level increases in the economy. Among the following effects which one will be observed?

- (a) Aggregate supply will increase because market competition becomes harsher.
- (b) Aggregate supply will increase because wages and production costs increase too.
- (c) Aggregate demand will fall because consumers will feel poorer.
- (d) Aggregate demand will fall because consumers anticipate a decrease in prices in the future.

Correct answer: (c)

Q26

Imagine that a small open economy faces an increase in exports. Among the following propositions which one is correct?

- (a) Even if the crowding out effect exactly compensates the effect of the Keynesian multiplier, the effect on income will still be positive.
- (b) Whatever the Keynesian multiplier and the crowding out effect are, the increase in exports has no impact on income as neither consumption nor investments are directly affected by exports.

- (c) Income will increase more than exports only if there is no crowding out effect.
- (d) If the crowding out effect is stronger than the Keynesian multiplier effect, income decreases.

Correct answer: (a)

Q27

Suppose that the government of Philliland decides to decrease public spending. Suppose also that unemployment is initially at its natural level. Among the following propositions which one is correct?

- (a) In the short-term, the new equilibrium moves on the left along the Phillips curve.
- (b) In the long-term, unemployment remains unchanged and only prices decrease.
- (c) In the long-term, unemployment increases and prices decrease.
- (d) The short-term Phillips curve moves upward, capturing a deterioration of the unemployment-inflation trade-off.

Correct answer: (b)

Q28

Among the following propositions regarding the natural rate of unemployment (i.e. NAIRU), which one(s) is(are) correct?

- i. The natural rate of unemployment is the unemployment rate which is compatible with a constantly increasing inflation rate.
 - ii. A disinflationary policy implemented by a credible central bank can rapidly reduce the natural unemployment rate.
 - iii. Only a shift to the left of the short term Phillips curve can cause a decrease in the natural unemployment rate.
- (a) None of the above propositions is correct.
 - (b) Only proposition i. is correct.
 - (c) Only proposition ii. is correct.
 - (d) Only propositions ii. and iii. are correct.

Correct answer: (a)

Q29

A shift to the left of the short-term aggregate supply function was observed during the Great Recession of 2008-2009, as well as during the Great Lockdown of 2020. Among the following propositions, which one can explain this phenomenon?

- (a) In both cases, a general feeling of pessimism caused a decrease in demand and thus in output.
- (b) In 2008, output had exceeded its full-employment level, and economic agents adjusted their price expectations upwards.
- (c) In 2020, central banks of countries affected by the pandemic adopted restrictive monetary policies to slow down production.
- (d) In 2008, aggregate supply shifted to the left because of disinflationary policies adopted by the US central bank (Fed), whereas in 2020, it shifted to the left because of the stop

of production due to the spread of coronavirus.

Correct answer: (b)

Q30

Among the following propositions regarding the efficiency of economic policies in a major economic crisis, which one is correct?

- (a) If the crisis was caused by a negative supply shock, by directly affecting aggregate supply, monetary policy is more efficient than fiscal policy.
- (b) High government debt may mitigate the positive effects of expansionary monetary policy because the central bank will not be able to reduce the interest rate below the rate attached to government bonds.
- (c) By reducing demand, global protectionist measures may reduce the effectiveness of fiscal policy.
- (d) In a liquidity trap situation, the central bank has no instrument at its disposal to increase the liquidity in circulation in the economy.

Correct answer: (c)